

Machine Learning Assignment 1: Polynomial Regression Analysis

Name: Hemant Jagnani | Roll Number: BT2024226 | GitHub: https://github.com/HemantJagnani/ML_Assignment1

1. Introduction

This report documents my end-to-end implementation and empirical evaluation of polynomial regression models for two assigned engineering problems. Each problem contains an independently generated calibration dataset and query test set.

Phase 1: Power Plant Steam Turbine Optimization (var1): The operational efficiency of a multi-stage geothermal steam turbine is governed by six input variables (x_1 to x_6), which represent percentage deviations for high-pressure steam valve adjustment, condenser coolant flow, pump hydraulic pressure, turbine blade pitch angle, non-condensable gas exhaust rate, and steam inlet pressure. The target variable y is the Net Power Score. The problem permits polynomial models up to degree 10.

Phase 2: Subterranean Thermal Reservoir Mapping (var2): To determine optimal drilling coordinates for new geothermal extraction wells, we map physical core samples taken across a 3D grid around the central basecamp landmark. The inputs (x_1, x_2, x_3) denote East-West offset, North-South offset, and vertical depth offset in meters. The target y is the Thermal Anomaly Score. The problem permits polynomial expansion up to degree 20. The primary goal is to select the optimal polynomial degree and regularization strength using validation evidence, preventing both underfitting and overfitting, and generating predictions for the held-out test datasets.

2. Data Inspection and Preprocessing

Before building model pipelines, I inspected all four CSV datasets assigned to my roll number (BT2024226). I checked row counts, column names, data types, missing values, infinite values, feature ranges, and target variance. Table 1 summarizes the structure and descriptive statistics of the provided datasets.

Dataset	Role	Rows	Features	Input Domain	Target Variable	Target Mean $\pm$ SD
BT2024226_train_var1.csv	Train	1,000	6 (x_1 – x_6)	$[-1.0, 1.0]$	Net Power Score	0.995 ± 3.180
BT2024226_test_var1.csv	Test	1,000	6 (x_1 – x_6)	$[-1.0, 1.0]$	Held-out Ground Truth	N/A
BT2024226_train_var2.csv	Train	1,000	3 (x_1 – x_3)	$[-1.0, 1.0]$	Thermal Anomaly	2.167 ± 6.707
BT2024226_test_var2.csv	Test	1,000	3 (x_1 – x_3)	$[-1.0, 1.0]$	Held-out Ground Truth	N/A

Table 1: Structural summary and descriptive statistics of the provided datasets.

Data Hygiene & Boundary Observations

All features in both problems are strictly continuous and bounded within $[-1.0, 1.0]$. There are zero NaN values, zero null entries, and zero infinite values. During duplicate inspection, I observed that **var2** training data contains exactly 8 repeated coordinate probes at spatial boundary vertices. For example, at the coordinate $(-1.0, 1.0, -1.0)$, four physical core readings yielded target values of 21.461, 21.680, 21.531, and 21.716, with a sample standard deviation of approximately 0.121. These repeated readings indicate local measurement variability, but are insufficient to estimate a global noise floor. The test datasets also contain repeated boundary points (10 in test_var1, 12 in test_var2), so test predictions must strictly preserve original row order.

Preprocessing & Leakage Prevention

Because all raw inputs are already normalized within $[-1.0, 1.0]$, raising variables to higher powers x^k naturally stays within $[-1.0, 1.0]$. This prevents the severe exponential numerical blowup that happens when unscaled features (> 1.0) are exponentiated. All modeling steps are encapsulated in scikit-learn Pipelines with *fit_intercept=True* and *include_bias=False* in PolynomialFeatures. No parameters are fit or inferred using the test set, guaranteeing zero data leakage.

3. Phase 1: Steam Turbine Optimization (var1)

Phase 1 has $D = 6$ operational variables. The number of polynomial features grows combinatorially as $P(D, d) = C(D + d, d) - 1$. While degree 1 has only 6 features, degree 5 expands to 461 features, and degree 10 expands to 8,007 features. Because the training set contains 1,000 samples while feature counts expand combinatorially, unregularized Ordinary Least Squares (OLS) becomes severely ill-conditioned and rank-deficient at higher degrees due to extreme collinearity. To ensure numerical stability and prevent unbounded coefficient variance, L2 Ridge regularization is used across all degree evaluations.

Degree	Features	Optimal α	5-Fold CV RMSE (Mean $\pm$ SD)	5-Fold CV R^2 (Mean $\pm$ SD)	Train RMSE	Train R^2
1	6	50.0	2.9683 $\pm$ 0.1408	0.1237 $\pm$ 0.0266	2.9512	0.1380
2	27	2.0	1.7445 $\pm$ 0.1082	0.6956 $\pm$ 0.0397	1.6831	0.7196
3	83	2.0	0.9953 $\pm$ 0.0807	0.9003 $\pm$ 0.0172	0.9013	0.9196
4	209	5.0	0.8749 $\pm$ 0.0669	0.9229 $\pm$ 0.0133	0.6798	0.9543
5 *	461	1.5	0.6787 $\pm$ 0.0424	0.9537 $\pm$ 0.0067	0.4194	0.9826
6	923	2.0	0.7330 $\pm$ 0.0371	0.9462 $\pm$ 0.0063	0.3788	0.9858
7	1,715	5.0	0.7184 $\pm$ 0.0380	0.9484 $\pm$ 0.0059	0.3902	0.9849
8	3,002	10.0	0.7672 $\pm$ 0.0257	0.9413 $\pm$ 0.0039	0.4128	0.9831
9	5,004	10.0	0.7831 $\pm$ 0.0390	0.9388 $\pm$ 0.0059	0.3860	0.9853
10	8,007	20.0	0.8248 $\pm$ 0.0279	0.9322 $\pm$ 0.0036	0.4275	0.9819

Table 2: Phase 1 (var1) 5-fold cross-validation results across degrees 1–10. (* indicates selected model)

Model Selection and Degree Justification

I evaluated degrees 1 through 10 using 5-fold cross-validation with a grid of Ridge regularization parameters α [0.001, 100.0]. As seen in Table 2, low degrees suffer from severe underfitting: degree 1 has an RMSE of 2.9683 ($R^2 = 0.1237$), and degree 2 has an RMSE of 1.7445 ($R^2 = 0.6956$). Validation error decreases rapidly up to degree 5, reaching a global minimum at **Degree 5** with **CV RMSE = 0.6787 $\pm$ 0.0424** and **CV $R^2 = 0.9537 \pm 0.0067$** at $\alpha = 1.5$. Moving from degree 5 to degree 6 doubles the feature count from 461 to 923, causing validation RMSE to rise to 0.7330. Degrees 7 through 10 continue to overfit, with validation RMSE degrading to 0.8248 at degree 10. Because Degree 5 achieves the lowest validation error with 461 polynomial features, Degree 5 ($\alpha = 1.5$) was chosen as the final model.

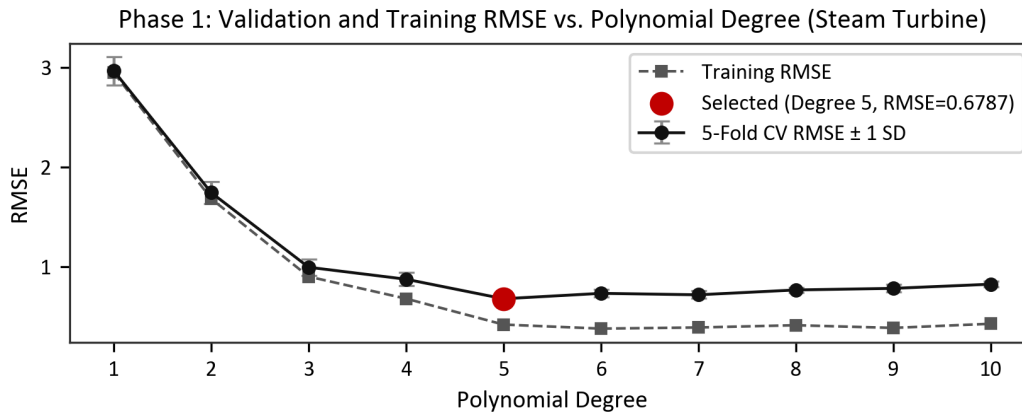


Figure 1: Phase 1 (var1) Validation and training RMSE curves across polynomial degrees 1 to 10.

4. Phase 2: Subterranean Thermal Reservoir Mapping (var2)

Phase 2 models a 3D geological heat map with $D = 3$ spatial offsets. Because the number of input variables is smaller, the polynomial feature count grows much more slowly: degree 1 has 3 features, degree 8 has 164 features, degree 12 has 454 features, and degree 20 has 1,770 features. All degrees from 1 to 20 were evaluated using 5-fold cross-validation. Table 3 lists the results for representative candidate degrees.

Degree	Features	Optimal α	5-Fold CV RMSE (Mean $\pm$ SD)	5-Fold CV R^2 (Mean $\pm$ SD)	Train RMSE	Train R^2
1	3	10.0	5.7286 $\pm$ 0.4915	0.2566 $\pm$ 0.0263	5.7176	0.2726
2	9	2.0	4.6645 $\pm$ 0.3635	0.5042 $\pm$ 0.0487	4.5813	0.5330
3	19	1.0	3.3814 $\pm$ 0.2615	0.7403 $\pm$ 0.0150	3.2255	0.7685
4	34	0.0 (OLS)	1.9230 $\pm$ 0.0964	0.9151 $\pm$ 0.0117	1.8026	0.9277
6	83	0.01	0.7937 $\pm$ 0.0390	0.9852 $\pm$ 0.0038	0.6852	0.9896
8	164	0.005	0.5024 $\pm$ 0.0192	0.9941 $\pm$ 0.0015	0.4160	0.9961
10	285	0.10	0.4909 $\pm$ 0.0179	0.9943 $\pm$ 0.0015	0.4151	0.9962
11	363	0.20	0.4945 $\pm$ 0.0140	0.9943 $\pm$ 0.0014	0.4163	0.9961
12 *	454	0.25	0.4880 $\pm$ 0.0145	0.9944 $\pm$ 0.0015	0.4130	0.9962
14	679	0.50	0.4936 $\pm$ 0.0118	0.9943 $\pm$ 0.0015	0.4160	0.9961
16	968	0.50	0.5030 $\pm$ 0.0089	0.9941 $\pm$ 0.0015	0.4122	0.9962
18	1,329	1.0	0.5150 $\pm$ 0.0094	0.9938	0.4219	0.9960
20	1,770	1.0	0.5260 $\pm$ 0.0102	0.9935 $\pm$ 0.0016	0.4190	0.9961

Table 3: Phase 2 (var2) 5-fold cross-validation results across degrees 1–20. (* indicates selected model)

Model Selection and Discussion

Low-degree polynomials severely underfit the complex 3D heat map: degree 1 has an RMSE of 5.7286, and degree 3 has an RMSE of 3.3814. As polynomial degree increases to degree 8, error drops dramatically to 0.5024 ($R^2 = 0.9941$). Between degrees 8 and 14, the validation curve enters a performance plateau where all models achieve $R^2 > 0.994$. The minimum mean CV RMSE across the entire search space occurs at **Degree 12** with **CV RMSE = 0.4880 $\pm$ 0.0145** and **CV $R^2 = 0.9944 \pm 0.0015$** at $\alpha = 0.25$. Although Degree 10 performs similarly with a CV RMSE of 0.4909, Degree 12 was retained because the model-selection criterion was the minimum mean cross-validation RMSE. Beyond degree 14, validation error climbs steadily toward 0.5260 at degree 20. Therefore, **Degree 12 ($\alpha = 0.25$)** was selected as the final model for Phase 2.

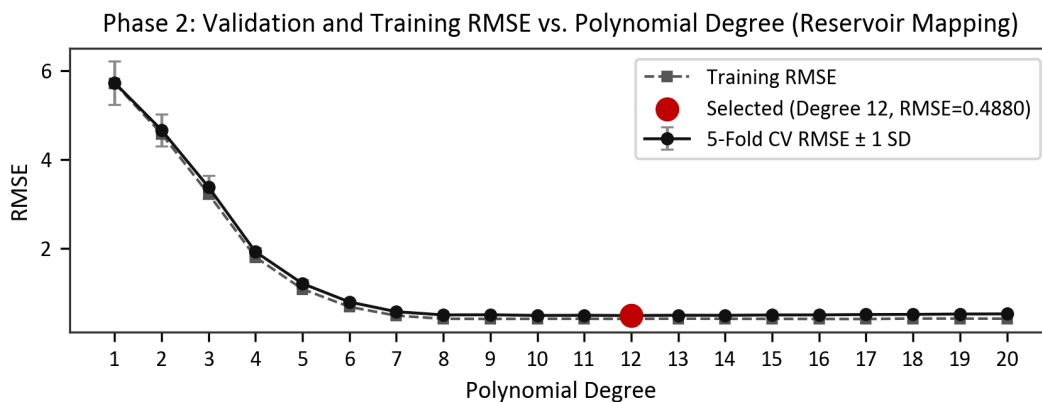


Figure 2: Phase 2 (var2) Validation and training RMSE curves across polynomial degrees 1 to 20.

5. Final Model Training and Test Inference

After selecting the optimal hyperparameters via cross-validation, I retrained each pipeline on 100% of the training dataset. Table 4 summarizes the final model parameters and full-dataset performance metrics.

Problem Phase	Selected Degree	Ridge α	Features	CV RMSE $\pm$ SD	CV $R^2 \pm$ SD	Train RMSE	Train R^2	Train MAE
Phase 1: Steam Turbine	5	1.50	461	0.6787 ± 0.0424	0.9537 ± 0.0067	0.4194	0.9826	0.3308
Phase 2: Reservoir Mapping	12	0.25	454	0.4880 ± 0.0145	0.9944 ± 0.0015	0.4130	0.9962	0.3334

Table 4: Summary of final model configurations and performance metrics.

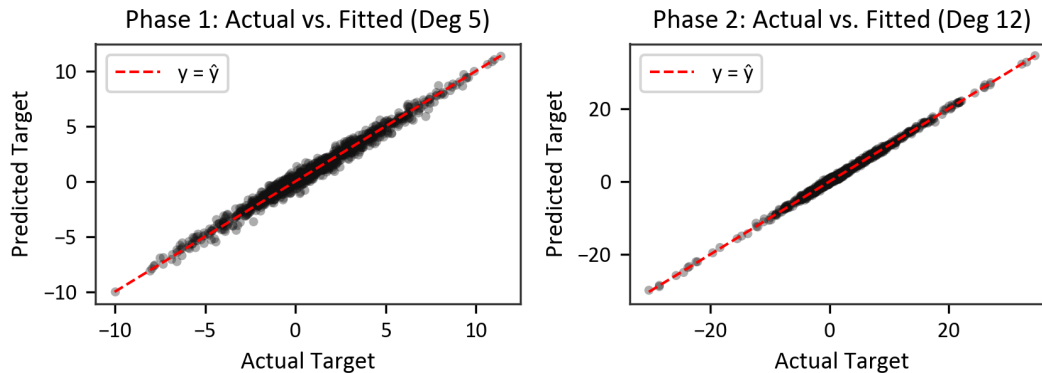


Figure 3: Actual target vs. fitted values for Phase 1 (left) and Phase 2 (right).

Test Predictions & Deliverable Verification

Using the fitted pipelines, I generated predictions for the test sets: **BT2024226_pred_var1.csv** and **BT2024226_pred_var2.csv**. Both files match the structure of *sample_submission.csv*: exactly 1,000 data rows, a single column named 'y', no index columns, and all finite values.

- **Var1 Test Predictions:** mean = 1.1599, std = 4.3395, min = -12.1471, max = 15.0941.
- **Var2 Test Predictions:** mean = 2.1647, std = 6.9393, min = -29.8759, max = 38.8158.

As instructed, test ground truth is held out for grading; no test scores were fabricated.

6. Technical Limitations and Practical Observations

1. **Collinearity of Monomial Basis:** Standard polynomial features $\{x_i^k\}$ become increasingly collinear at higher degrees, causing the design matrix condition number to grow rapidly. While L2 Ridge penalty stabilizes the inversion, using orthogonal polynomials (such as Chebyshev or Legendre polynomials) would provide better numerical stability.
2. **Boundary Behavior of High-Degree Polynomial Models:** High-degree polynomial representations can exhibit increased sensitivity near domain boundaries ($|x| \approx 1.0$), where prediction behavior may be more variable than in the interior of the domain.
3. **Measurement Variability:** Repeated observations at identical spatial coordinates exhibit local variability (sample SD ≈ 0.121). While these differences reflect measurement uncertainty in physical sampling, they are restricted to isolated boundary probes and do not represent a characterized global noise floor across the entire domain.

7. Conclusion

For student roll number **BT2024226**, I selected **Degree 5 Ridge ($\alpha = 1.5$)** for Phase 1 (Steam Turbine Optimization), achieving a 5-fold CV RMSE of 0.6787 and CV R^2 of 0.9537. For Phase 2 (Subterranean Reservoir Mapping), I selected **Degree 12 Ridge ($\alpha = 0.25$)**, achieving a 5-fold CV RMSE of 0.4880 and CV R^2 of 0.9944. All predictions and code deliverables have been generated and verified against the assignment requirements.